Class #9 - The Power Method

1. Assume A is real $n \times n$ and diagonalizable, so it has _____linearly independent eigenvectors $\{\mathbf{v_1}, \cdots, \mathbf{v_n}\}$ in _____. Suppose its corresponding _____eigenvalues are so that

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots.$$

Then, for any $\mathbf{x}$ in $\mathbf{R^n}$,

$$A^k\mathbf{x} = _____.$$

Assume $c_1 \neq 0$, then

Hence, for large k, the _____$A^k\mathbf{x}$ points almost in the same direction as _____or _____.

We can scale $A^k\mathbf{x}$ for each k so that the vector is not too long in its magnitude.

2. Power Method:
(Step 1) Choose $\mathbf{x_0}$.
(Step 2) For $k = 1, 2, 3, \cdots$ do until convergence

$$\hat{\mathbf{x_k}} = A\mathbf{x_{k-1}}$$

$$\mathbf{x_k} = \hat{\mathbf{x_k}}/\max(\hat{\mathbf{x_k}})$$

3. Under the conditions: $c_1 \neq 0$, A diagonalizable, λ_1 is the only dominant eigenvalue, $\max(\hat{\mathbf{x_k}}) \to \lambda_1$, and $\{\mathbf{x_k}\} \to \mathbf{w_1}$, a multiple of $\mathbf{v_1}$, as $k \to \infty$.

Since

$$\mathbf{x_k} = \frac{A^k}{\max(A^k \mathbf{x_0})},$$

by the same discussion as in #1,

$$\mathbf{x_k} = \frac{A^k}{\max(A^k \mathbf{x_0})} \to \alpha \mathbf{v_1}, \quad \text{and} \quad \max(\hat{\mathbf{x_k}}) \to \lambda_1.$$

4. Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$, $\mathbf{x_0} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. eigenvalues: $0, -0.6235, 9.6235$. So, $\lambda_1 = 9.6235$, and $\mathbf{v_1} = \begin{bmatrix} 0.3851 \\ 0.5595 \\ 0.7339 \end{bmatrix}$.

$k = 1$: $\hat{\mathbf{x_1}} =$ _____ $= \begin{bmatrix} 6 \\ 9 \\ 12 \end{bmatrix}$, $\max(\hat{\mathbf{x_1}}) =$ _____, $\mathbf{x_1} =$ _____.

$k = 2$: $\hat{\mathbf{x_2}} =$ _____ $= \begin{bmatrix} 5.00 \\ 7.25 \\ 9.50 \end{bmatrix}$, $\max(\hat{\mathbf{x_2}}) =$ _____, $\mathbf{x_2} = \begin{bmatrix} 0.5263 \\ 0.7632 \\ 1.0000 \end{bmatrix}$.

$k = 3$: $\hat{\mathbf{x_3}} =$ _____ $= \begin{bmatrix} 5.0526 \\ 7.3421 \\ 9.6316 \end{bmatrix}$, $\max(\hat{\mathbf{x_3}}) =$ _____, $\mathbf{x_3} = \begin{bmatrix} 0.5246 \\ 0.7623 \\ 1.0000 \end{bmatrix}$.